Inflammation, Amyloid Burden, and Cognitive Decline in aMCI

Data Analysis Plan and Sample Size Justification

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Background & Study Aims

Scientific Context

- Peripheral inflammation may accelerate cognitive decline in aMCI
- Amyloid burden may amplify the inflammation–cognition relationship
- Combined n = 175 (125 aMCI + 50 HC); one-year follow-up

Aim 1

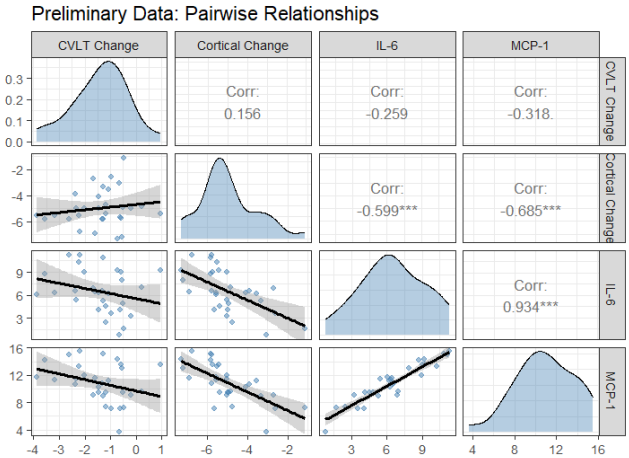
Longitudinal associations between four inflammatory markers (IL-6, TNF-α, MCP-1, Eotaxin-1) and change in six cognitive/cortical outcomes

Aim 2

Does amyloid burden (continuous SUVR) *moderate* the inflammation–cognition association?
(aMCI only, n = 125)

Outcomes: CVLT 20′ delay, CVLT Recognition D′, Benson 15′ delay, MST, Story Recall, AD-signature cortical thickness change

Preliminary Data: Pairwise Relationships (n = 30)



- Complete data
- Inflammatory markers positively skewed
- **IL-6 and MCP-1 strongly correlated** ($r = 0.934$) $\Rightarrow$ multicollinearity concern for joint models
- Both markers negatively associated with CVLT and cortical thickness change
- Cortical thickness shows stronger associations ($r = -0.60$ to -0.69) than memory ($r = -0.26$ to -0.32)

Key takeaway: Marginal correlations $\times$ 0.75 shrinkage $\Rightarrow$ assumed partial correlations for power

Aim 1: Analysis Plan & Power

Model Structure

outcome change $\sim$ marker + baseline + covariates

- 24 single-marker models (4×6)
- Baseline outcome always included (ANCOVA)
- Age + sex fixed; APOE, BMI, BP, etc. evaluated as confounders/precision variables
- Joint 4-marker model; PCA if VIF > 10

Type I Error

Tests not independent (correlated markers & outcomes)
Primary: $\alpha = 0.05$ **Sensitivity:** $\alpha = 0.025$
Emphasis on consistent patterns across outcomes/markers

Table 1. Selected Aim 1 power ($N = 175$, $p = 8$, $q = 1$, $v = 165$).

Marker	Outcome	|pc|	.05	.025
IL-6	CVLT 20′	0.19	0.721	0.618
MCP-1	CVLT 20′	0.24	0.902	0.842
TNF-α	MST	0.14	0.460	0.350
IL-6	Cortical	0.45	1.000	1.000
MCP-1	Cortical	0.51	1.000	1.000
Eotaxin-1	Cortical	0.34	0.997	0.994
MDE |pc| at 80% power:			0.208 / 0.247	

- Cortical Δ: power ≥ 0.997 for all markers
- MCP-1 best powered for memory
- TNF-α / Eotaxin-1 underpowered for small memory effects

Aim 2: Analysis Plan & Type I Error

Model Structure (N = 125 aMCI)

outcome change $\sim$ marker + SUVR + marker $\times$ SUVR + baseline + covariates

- Primary test: **interaction term** ($\alpha = 0.05$)
- Continuous SUVR in primary analysis

Power Method

Interaction = **difference in correlations** between amyloid+ and amyloid− groups

$\Rightarrow$ `corr.2samp()` (Fisher's z-test)

Assuming 40% amyloid prevalence:
$n_1 = 50$ (amyloid+), $n_2 = 75$ (amyloid−)

No Formal Alpha Correction

Correction would make detection infeasible.

Confidence requires **both**:

(1) Consistent direction across outcomes & markers

(2) Biologically plausible direction: higher inflammation + higher amyloid $\Rightarrow$ *worse* cognitive outcomes

No preliminary amyloid data

Power evaluated across plausible ρ_+ and ρ_- scenario ranges Min detectable $\Delta\rho$ at 80% power

Outcome	ρ_-	$\Delta\rho$ needed
Memory	−0.15	0.436
Cortical	−0.35	0.359

Aim 2: Power Results by Scenario

Table 2. Aim 2 power (40/60 split, $n_1 = 50$, $n_2 = 75$, $\alpha = 0.05$).

Scenario	Outcome	ρ_+	ρ_-	$\Delta\rho$	Power
Small	Memory	−0.35	−0.15	0.20	0.211
Moderate	Memory	−0.50	−0.15	0.35	0.573
Large	Memory	−0.65	−0.10	0.55	**0.953**
Small	Cortical	−0.55	−0.35	0.20	0.276
Moderate	Cortical	−0.70	−0.35	0.35	0.771
Large	Cortical	−0.80	−0.20	0.60	**0.998**

- Adequate power **only for large** interactions
- Cortical Δ better powered than memory at moderate effects
- Results consistent across amyloid prevalence assumptions

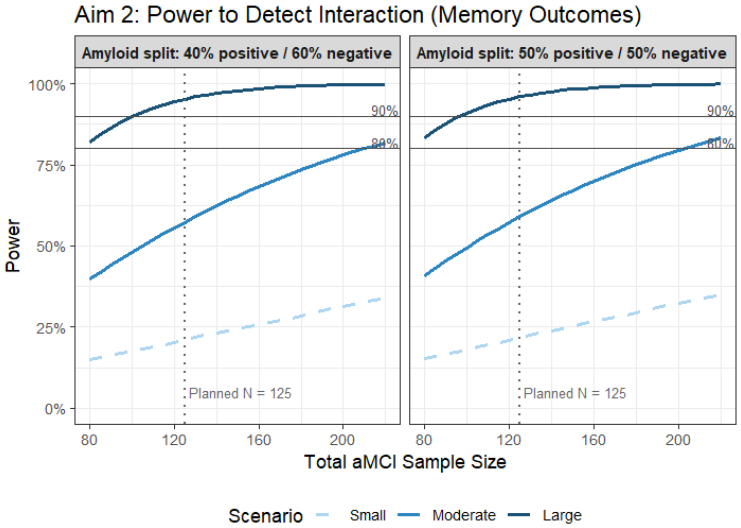
Table 3. Sensitivity across amyloid splits.

Scenario	Outcome	$\Delta\rho$	30/70	40/60	50/50
Large	Memory	0.55	0.924	0.953	0.960
Large	Cortical	0.60	0.994	0.998	0.998
Moderate	Memory	0.35	0.519	0.573	0.590
Moderate	Cortical	0.35	0.716	0.771	0.786
Small	Memory	0.20	0.191	0.211	0.217
Small	Cortical	0.20	0.250	0.276	0.284

Limitation

Study underpowered for small-to-moderate interactions. Null memory findings should be interpreted with caution.

Aim 2: Power Curves Across Sample Sizes



- Planned N = 125 marked with vertical dotted line
- **Large** scenario exceeds 80% at N = 125 under both splits
- **Moderate** scenario remains below 80% even at N = 220
- Balanced split (50/50) provides slight power advantage over 40/60
- Would need N ≈ 210+ for 80% power at moderate memory interactions

R v4.3.3, `powertools` v1.0.0